MATH INQUIRERS HOLIDAY PROGRAM

Weekly Email Pack

Parent and tutor emails for the five weeks of the P4 Ages 9-11 holiday program

Notice it. Test it. Explain why.

How to use this pack

- Send the parent email and the Week N Student Pages PDF the day before the first lesson of that week.
- Send the tutor email with the Week N Tutor Scripts and Answers PDF at least two days before. Tutors only; the answers must not go to parents.
- Put parents in BCC so no one sees another family's email address. Get parents' consent to email before you start (PDPA).
- Replace the words in [square brackets] before sending.
- The PDFs are in the weekly folder (Math_Inquirers_P4_Weekly_PDFs.zip): Week1_Student_Pages.pdf to Week5_Student_Pages.pdf and Week1_Tutor_Scripts_and_Answers.pdf to Week5_Tutor_Scripts_and_Answers.pdf. The Week 1 student PDF also holds the inquiry journal page and all the cut-outs; the Week 5 PDF ends with the certificate.
- The slide deck links in the tutor emails are private to you. Share each deck with the tutor first, or send them the file.

Week 1: Lessons 1 and 2

Parent email

Subject: Math Inquirers P4 Ages 9-11 Week 1: Factors and multiples and Numbers to 100 000

Attachment: Week1_Student_Pages.pdf

Dear parents,

This week our Math Inquirers (Ages 9-11 (builds on Primary 4 maths)) meet two topics, in lessons 1 and 2.

Lesson 1, Factor Hunters (Factors and multiples). Factors and multiples: listing factor pairs in order, common factors and common multiples, and square numbers.

Lesson 2, Big Number Puzzles (Numbers to 100 000). Numbers to 100 000: rounding, estimating, and riddles about 5-digit numbers.

Your child's workbook pages for this week are attached. Please print them single-sided if you can. Children will work on the pages in class, so there is no need to finish them in advance.

The cut-out pages are at the end of the Week 1 PDF (Fraction wall, square grid, dot grid, cube net, right-angle testers). Please cut them out and keep them in one envelope; we use them through the program. The inquiry journal page is at the front.

One thing to try at home (5 minutes): Use 24 coins or buttons. How many different equal rows can your child make? Which numbers have the most rows? Also try: Is 45 a multiple of 4? (No: 44 and 48 are); 18 as a product, 3 ways? (1 × 18, 2 × 9, 3 × 6).

Ask your child: "How do you know you have found every factor?" or "Which digit decides how you round?"

We are not looking for speed. We are looking for a child who can say what they noticed, test it, and explain why. If your child is stuck, please ask "What do you notice?" before you give the answer.

Warm regards,
[Your name]
Preciouslyours Pte Ltd
[Contact number]

Week 1: Tutor email

Subject: Math Inquirers P4 Ages 9-11 Week 1 tutor pack: Lessons 1 and 2

Attachments: Week1_Tutor_Scripts_and_Answers.pdf, Week1_Student_Pages.pdf

Hi [tutor name],

Here is your pack for week 1: Factor Hunters; Big Number Puzzles.

Attached are the minute-by-minute scripts with answers for both lessons, and the student pages. The slide decks are linked below.

- Lesson 1 deck: https://claude.ai/artifact/NNqtsRzk1wUPXUyyaDffmq
- Lesson 2 deck: https://claude.ai/artifact/QguJR6ubuHSNx1r2FaJNEQ

Before the lesson, please:

1. Read both scripts once and try the Go further questions yourself.
2. Check your materials are ready. Lesson 1: 36 square tiles or squared paper per pair, mini whiteboards, coloured pencils, a 1 to 50 number strip. Lesson 2: Digit cards 0 to 9 (one set per child: cut them from the Digit, point, sign and bracket cards page), mini whiteboards, a place-value chart (ten thousands to ones), a blank number line strip.
3. After each lesson, tick the Readiness Tracker for each child (four quick ticks) and send me the sheet.

Please do not forward the answer pages to parents.

Thank you,
[Your name]

Week 2: Lessons 3 and 4

Parent email

Subject: Math Inquirers P4 Ages 9-11 Week 2: Four operations and Fractions

Attachment: Week2_Student_Pages.pdf

Dear parents,

This week our Math Inquirers (Ages 9-11 (builds on Primary 4 maths)) meet two topics, in lessons 3 and 4.

Lesson 3, Operation Detectives (Four operations). Four operations: order of operations, brackets, missing signs and bar models.

Lesson 4, Fraction Frontier (Fractions). Fractions: improper fractions and mixed numbers, comparing fractions, and fractions of a quantity with bar models.

Your child's workbook pages for this week are attached. Please print them single-sided if you can. Children will work on the pages in class, so there is no need to finish them in advance.

Please keep the Week 1 envelope handy. This week's lessons use: fraction wall.

One thing to try at home (5 minutes): Write 6 + 4 × 3 and ask for the answer. Then add brackets in different places and see how the answer changes. Also try: 1 000 − 368 = ? (632); 4 × 20 = ? (80).

Ask your child: "Which do you do first, and why?" or "Can you show it with a bar?"

We are not looking for speed. We are looking for a child who can say what they noticed, test it, and explain why. If your child is stuck, please ask "What do you notice?" before you give the answer.

Warm regards,
[Your name]
Preciouslyours Pte Ltd
[Contact number]

Week 2: Tutor email

Subject: Math Inquirers P4 Ages 9-11 Week 2 tutor pack: Lessons 3 and 4

Attachments: Week2_Tutor_Scripts_and_Answers.pdf, Week2_Student_Pages.pdf

Hi [tutor name],

Here is your pack for week 2: Operation Detectives; Fraction Frontier.

Attached are the minute-by-minute scripts with answers for both lessons, and the student pages. The slide decks are linked below.

- Lesson 3 deck: https://claude.ai/artifact/RGNUJdPd4XxCRx4rhAeVTv
- Lesson 4 deck: https://claude.ai/artifact/GDpF53SmdUNoYKcdmUHdvP

Before the lesson, please:

1. Read both scripts once and try the Go further questions yourself.
2. Check your materials are ready. Lesson 3: Mini whiteboards and markers; number cards 0 to 9, sign cards +, −, ×, ÷ and two pairs of bracket cards (all on the Digit, point, sign and bracket cards page, one page per child); scrap paper for bar models. Lesson 4: Paper strips (one per child, to fold into halves, quarters and eighths), the three pizza circles from the cut-out page (to cut into quarters), 30 counters, mini whiteboards and markers.
3. After each lesson, tick the Readiness Tracker for each child (four quick ticks) and send me the sheet.

Please do not forward the answer pages to parents.

Thank you,
[Your name]

Week 3: Lessons 5 and 6

Parent email

Subject: Math Inquirers P4 Ages 9-11 Week 3: Decimals and Angles

Attachment: Week3_Student_Pages.pdf

Dear parents,

This week our Math Inquirers (Ages 9-11 (builds on Primary 4 maths)) meet two topics, in lessons 5 and 6.

Lesson 5, Decimal Detectives (Decimals). Decimals: ordering, rounding, and decimals in money and measures.

Lesson 6, Angle Turns (Angles). Angles: measuring with a protractor, estimating, and turns on a compass or clock.

Your child's workbook pages for this week are attached. Please print them single-sided if you can. Children will work on the pages in class, so there is no need to finish them in advance.

Please keep the Week 1 envelope handy. This week's lessons use: right-angle testers.

One thing to try at home (5 minutes): Look at a supermarket receipt. Order the prices from smallest to largest and round the total to the nearest dollar. Also try: Round 4.68 to 1 decimal place. (4.7); How much more than $3.05 is $5? ($1.95).

Ask your child: "What is each digit worth?" or "Is it more or less than a right angle?"

We are not looking for speed. We are looking for a child who can say what they noticed, test it, and explain why. If your child is stuck, please ask "What do you notice?" before you give the answer.

Warm regards,
[Your name]
Preciouslyours Pte Ltd
[Contact number]

Week 3: Tutor email

Subject: Math Inquirers P4 Ages 9-11 Week 3 tutor pack: Lessons 5 and 6

Attachments: Week3_Tutor_Scripts_and_Answers.pdf, Week3_Student_Pages.pdf

Hi [tutor name],

Here is your pack for week 3: Decimal Detectives; Angle Turns.

Attached are the minute-by-minute scripts with answers for both lessons, and the student pages. The slide decks are linked below.

- Lesson 5 deck: https://claude.ai/artifact/YB4HP2GbXMT1sDi3BWpq9x
- Lesson 6 deck: https://claude.ai/artifact/3UhJSwZ2Zh6rddxMC5GLT1

Before the lesson, please:

1. Read both scripts once and try the Go further questions yourself.
2. Check your materials are ready. Lesson 5: Digit cards 0 to 9 and a decimal point card for each child (cut from the Digit, point, sign and bracket cards page), a place-value chart (ones, tenths, hundredths, thousandths), mini whiteboards, a metre ruler, play money. Lesson 6: A clear 180° protractor and a 30 cm ruler for each child (a plastic school protractor is fine, cheap to buy; bring from home; the tutor keeps 2 or 3 spares). The paper clock from the cut-out page with a split pin (or a real clock), a square of paper for checking right angles, mini whiteboards, a compass card on the floor or wall.
3. After each lesson, tick the Readiness Tracker for each child (four quick ticks) and send me the sheet.

Please do not forward the answer pages to parents.

Thank you,
[Your name]

Week 4: Lessons 7 and 8

Parent email

Subject: Math Inquirers P4 Ages 9-11 Week 4: Area and Perimeter and Symmetry and Nets

Attachment: Week4_Student_Pages.pdf

Dear parents,

This week our Math Inquirers (Ages 9-11 (builds on Primary 4 maths)) meet two topics, in lessons 7 and 8.

Lesson 7, Area Architects II (Area and Perimeter). Area and perimeter: figures made of rectangles, missing lengths, and same area with different perimeters.

Lesson 8, Symmetry and Nets (Symmetry and Nets). Symmetry and nets: finishing symmetric figures, lines of symmetry, and nets that fold into a cube.

Your child's workbook pages for this week are attached. Please print them single-sided if you can. Children will work on the pages in class, so there is no need to finish them in advance.

Please keep the Week 1 envelope handy. This week's lessons use: square grid, dot grid, cube net.

One thing to try at home (5 minutes): Draw a room plan on squared paper. Ask for the area of the floor and the length of the skirting around it. Also try: Perimeter of a rectangle 12 cm by 5 cm? (34 cm); A square has perimeter 28 cm. Its area? (49 cm^2).

Ask your child: "Did you find all the missing lengths first?" or "Which faces are opposite each other?"

We are not looking for speed. We are looking for a child who can say what they noticed, test it, and explain why. If your child is stuck, please ask "What do you notice?" before you give the answer.

Warm regards,
[Your name]
Preciouslyours Pte Ltd
[Contact number]

Week 4: Tutor email

Subject: Math Inquirers P4 Ages 9-11 Week 4 tutor pack: Lessons 7 and 8

Attachments: Week4_Tutor_Scripts_and_Answers.pdf, Week4_Student_Pages.pdf

Hi [tutor name],

Here is your pack for week 4: Area Architects II; Symmetry and Nets.

Attached are the minute-by-minute scripts with answers for both lessons, and the student pages. The slide decks are linked below.

- Lesson 7 deck: https://claude.ai/artifact/4FHm4cmWU9MLhWRqGimYw1
- Lesson 8 deck: https://claude.ai/artifact/5GwTmyq7STRgRWEUeSbHej

Before the lesson, please:

1. Read both scripts once and try the Go further questions yourself.
2. Check your materials are ready. Lesson 7: 1 cm square tiles or 24 paper squares per child (cut the Square grid page into squares), 1 cm grid paper, scissors, a 30 cm ruler, mini whiteboards, a paper strip 8 squares by 2 squares for cutting. Lesson 8: Squared paper (2 cm squares), scissors, sticky tape, 6 paper squares per child (cut from the Square grid page), a dice, a small mirror (optional), mini whiteboards, coloured pencils.
3. After each lesson, tick the Readiness Tracker for each child (four quick ticks) and send me the sheet.

Please do not forward the answer pages to parents.

Thank you,
[Your name]

Week 5: Lessons 9 and 10

Parent email

Subject: Math Inquirers P4 Ages 9-11 Week 5: Tables and line graphs and Mixed heuristics

Attachment: Week5_Student_Pages.pdf

Dear parents,

This week our Math Inquirers (Ages 9-11 (builds on Primary 4 maths)) meet two topics, in lessons 9 and 10.

Lesson 9, Data Detectives (Tables and line graphs). Tables and line graphs: reading trends, finding missing values, and spotting misleading scales.

Lesson 10, Challenge Day (Mixed heuristics). Choosing a strategy: guess and check, supposition, working backwards and logic grids.

Your child's workbook pages for this week are attached. Please print them single-sided if you can. Children will work on the pages in class, so there is no need to finish them in advance.

No cut-outs are needed this week. Please keep the Week 1 envelope handy. The Certificate of Completion is the last page of the Week 5 PDF.

One thing to try at home (5 minutes): Track the temperature or your child's steps each day for a week and draw a line graph together. Also try: Four numbers total 200. Three are 45, 60 and 50. The fourth? (45); Bigger change: 120 to 95, or 60 to 90? (60 to 90 (30)).

Ask your child: "What does the steepest part tell us?" or "How did you start? What would you try next time?"

We are not looking for speed. We are looking for a child who can say what they noticed, test it, and explain why. If your child is stuck, please ask "What do you notice?" before you give the answer.

Warm regards,
[Your name]
Preciouslyours Pte Ltd
[Contact number]

Week 5: Tutor email

Subject: Math Inquirers P4 Ages 9-11 Week 5 tutor pack: Lessons 9 and 10

Attachments: Week5_Tutor_Scripts_and_Answers.pdf, Week5_Student_Pages.pdf

Hi [tutor name],

Here is your pack for week 5: Data Detectives; Challenge Day.

Attached are the minute-by-minute scripts with answers for both lessons, and the student pages. The slide decks are linked below.

- Lesson 9 deck: https://claude.ai/artifact/Nh7jmRZLL4aGgv5tFb2xuT
- Lesson 10 deck: https://claude.ai/artifact/JBssFj1qkvX7H1acFxmSsG

Before the lesson, please:

1. Read both scripts once and try the Go further questions yourself.
2. Check your materials are ready. Lesson 9: Mini whiteboards, a ruler and two coloured pencils per child, the blank graph grids from the kit page (or squared paper), the queue table for the explore task. Lesson 10: Four puzzle cards (A to D) for the explore stations (cards A and B live; C and D take-home), mini whiteboards, counters or cubes, certificates, stamps.
3. After each lesson, tick the Readiness Tracker for each child (four quick ticks) and send me the sheet.

Please do not forward the answer pages to parents.

Thank you,
[Your name]

Week 5 closing email to parents

Subject: Math Inquirers P4 Ages 9-11: thank you, and your child's certificate

Dear parents,

Thank you for joining us for the Math Inquirers Holiday Program (Ages 9-11 (builds on Primary 4 maths)). Your child's Certificate of Completion is the last page of the Week 5 PDF. Please print it and add your child's name.

We would love your feedback in two minutes: [feedback form link]. What did your child enjoy most? What should we change?

If your child enjoyed the puzzles, tell us. Think harder and Go further questions are open to every child, and we will share more puzzles to try in the next holiday.

Warm regards,
[Your name]